

Homework 9 Solutions

Donald Timpson

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Problem 20.1

- a) If x^2 is not odd, then x is not odd.
- b) If $2^p - 2$ is not divisible by p , then p is not prime.
- c) If x^2 is not positive, then x is zero.
- d) If the parallelogram is not a rhombus, then the diagonals of a parallelogram are not perpendicular.
- e) If the car will not start, then the battery is not fully charged.
- f) If $\neg C$ then $\neg A$ and $\neg B$.

Problem 20.2

The contrapositive of the contrapositive is just the original if-then statement.

Problem 20.3

Since $\neg A \Rightarrow \neg B$ is logically equivalent to $B \Rightarrow A$, then $A \Rightarrow B$ and $\neg A \Rightarrow \neg B$ is the same as A iff B .

Problem 20.5

Proof. Consecutive integers cannot both be even.

1. Let x and y be consecutive integers such that both x and y are even.
2. Then $\exists k_0, k_1 \in \mathbb{Z}$ such that $x = 2k_0$ and $y = 2k_1$.
3. Observe that $y = x + 1$. (Definition of consecutive)
4. Then $2k_1 = 2k_0 + 1 \Rightarrow 2(k_1 - k_0) = 1 \Rightarrow k_1 - k_0 = \frac{1}{2}$.
5. Since the integer set is closed under addition, then $k_1 + (-k_0) = \frac{1}{2}$ and k_1 and k_0 both being integers is a contradiction.
6. Therefore consecutive integers cannot both be even. □

Problem 20.6

Proof. Consecutive integers cannot both be odd.

1. Let x and $x + 1$ be consecutive integers such that they are both odd.
2. Then $\exists a, b \in \mathbb{Z}$ such that $x = 2a + 1$ and $x + 1 = 2b + 1$.
3. Then $x + 1 = 2b + 1$ gives $x = 2b$.
4. Then $x = 2b$ gives $2|x$ gives x is even.
5. Then x is even and odd which contradicts proposition 20.3.
6. Therefore consecutive integers cannot both be even. □

Problem 20.7

Proof. If the sum of two primes is prime, then one of the primes must be 2.

1. Let p_0 and p_1 be prime numbers such that $p_0 + p_1$ is prime, and $p_0 \neq 2$ and $p_1 \neq 2$.
2. Since $p_0 \neq 2$ and $p_1 \neq 2$, and since 2 is the only even prime, then p_0 and p_1 must be odd.
3. Then $\exists k_0, k_1 \in \mathbb{Z}$ such that $p_0 = 2k_0 + 1$ and $p_1 = 2k_1 + 1$.
4. Then $p_0 + p_1 = 2k_0 + 1 + 2k_1 + 1 = 2(k_0 + k_1 + 1)$.
5. Since $\exists k \in \mathbb{Z}$ such that $p_0 + p_1 = 2k$, namely $k = k_0 + k_1 + 1$, then $2|(p_0 + p_1)$.
6. Since 2 is the smallest prime, then $2 < p_0 + p_1$.
7. Therefore $p_0 + p_1$ is divisible by some number greater than 1 and less than itself.
8. Therefore if the sum of two primes is prime, then one of the primes must be 2. □

Problem 20.10

Proof. If $1 < a$ then $1 < \sqrt{a} < a$.

1. Let $a \in \mathbb{R}$ such that $a > 1$ and let $x = \sqrt{a}$.
2. Assume by way of contradiction that $x \leq 1$.
3. Then $x \cdot x \leq 1 \cdot x \leq 1 \cdot 1$ which gives $a \leq 1 \Rightarrow \Leftarrow$.
4. Therefore $1 < x$.
5. Now assume by way of contradiction that $a \leq x$.
6. Since $1 < a$ and $a \leq x$ then $1 \cdot a < a \cdot x$.
7. Again, since $1 \cdot a < a \cdot x$ and $a \leq x$ then $1 \cdot a \cdot a < a \cdot x \cdot x$.
8. Then $1 \cdot a \cdot a < a \cdot x \cdot x$ gives $a^2 < a^2 \Rightarrow \Leftarrow$.
9. Therefore $x < a$.
10. Therefore $1 < \sqrt{a} < a$. □